

5 (long)

A pin jointed steel truss structure is shown in Fig. 5. The truss is subject to a vertical point load of 10 kN applied at joint D. The members of the truss have cross-sectional area A and Youngs Modulus E such that $EA = 10^7$ N.

(a) Determine the forces in each bar of the truss. [10]

(b) Determine the displacement of joint B. [20]

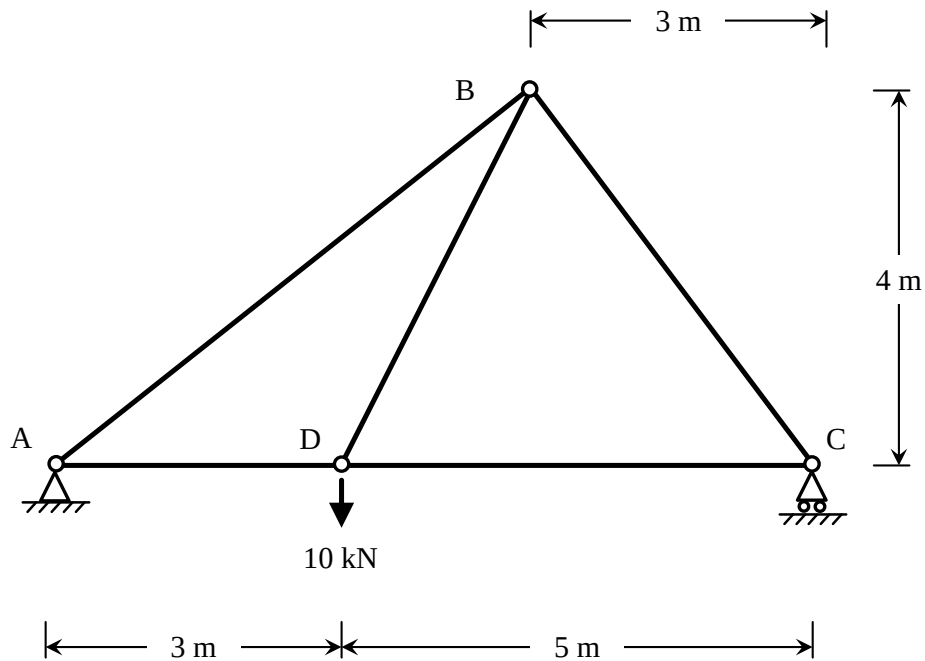


Fig. 5